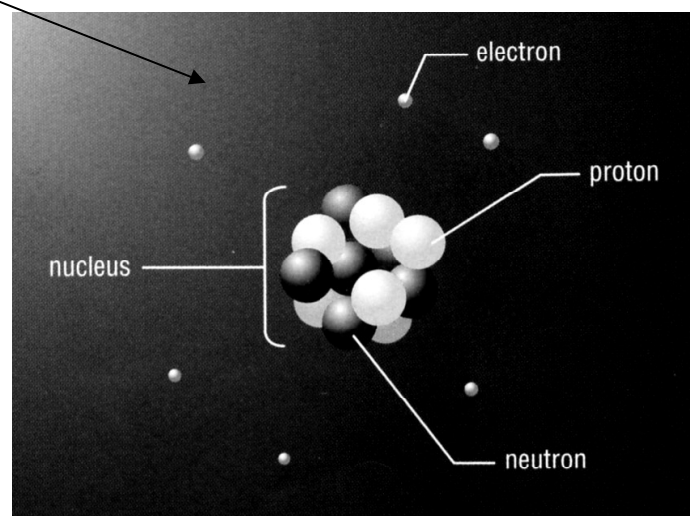
STATIC ELECTRICITY

Atomic Structure

The structure of the atom has a cloud of electrons orbiting around a nucleus in energy shells.

It is the ***outer electrons that are involved in the production of static electricity***.



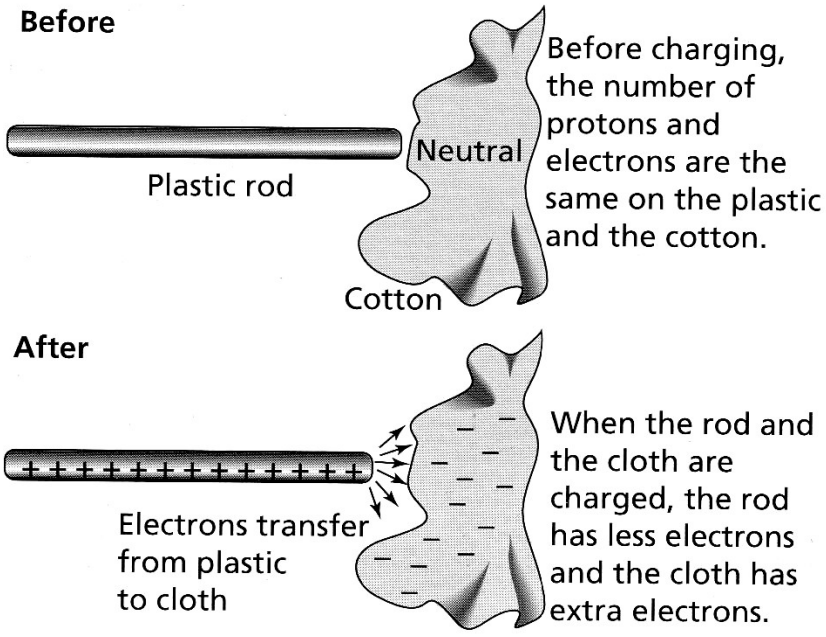
Static Electricity

Friction between two surfaces can rub electrons off one object and onto another.

The object that ***gains electrons becomes negative*** - the other becomes positive. This produces static electricity.

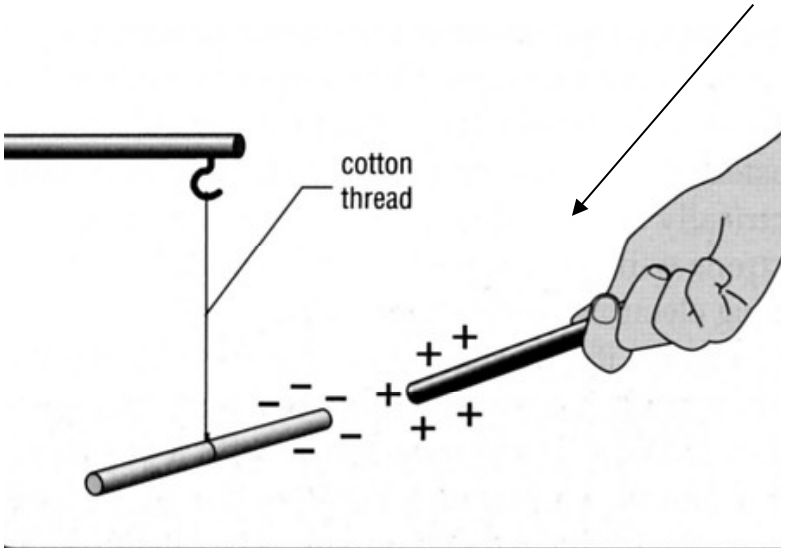
Static Electricity: stationary charges on an object.

e.g. A plastic comb rubbed on a cotton cloth becomes positively charged by losing electrons.



Interaction of Charges

Charge Law: ***Similar charges repel*** each other while ***opposite charges attract*** each other.



Examples of Static Electricity

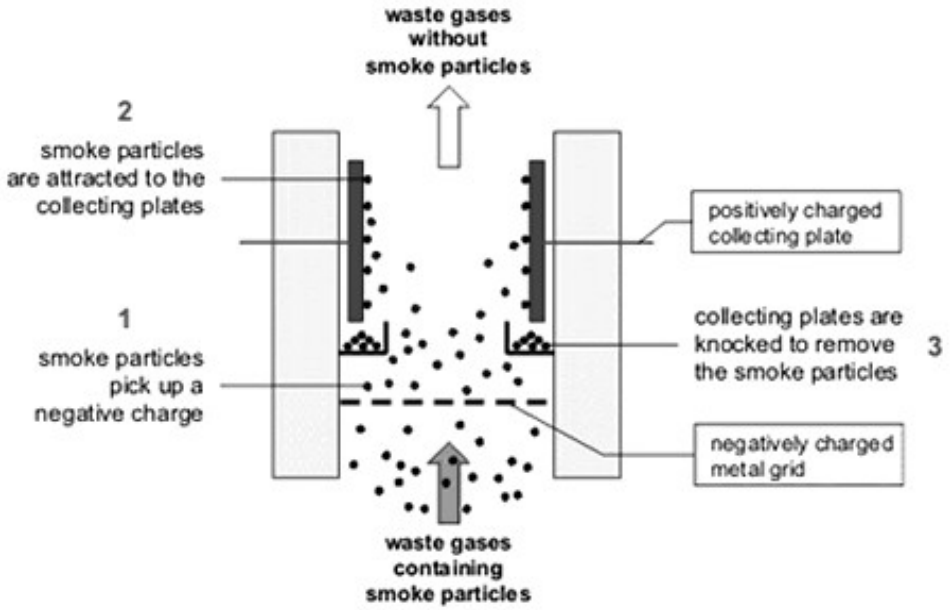
Lightning



- Caused by the friction between the air and the clouds.
- Convection currents within the cloud also cause ice particles to produce static charge.
- Charge on the ground below the ground becomes oppositely charged.
- Lightning is discharged to the ground or tallest structure with a large charge density.

Pollution Control

- Chimneys have special ***electrostatic traps*** to attract smoke particles.
- This reduces the amount reaching the atmosphere.



Getting "Zapped"

- Moving cars build up a charge as they move through the air.
- Passengers sometimes get "zapped" as they get out of the car and touch the metal part of the door.
- The charged flows around the person to the Earth.